Wine Quality Prediction

Regression from physicochemical measurements

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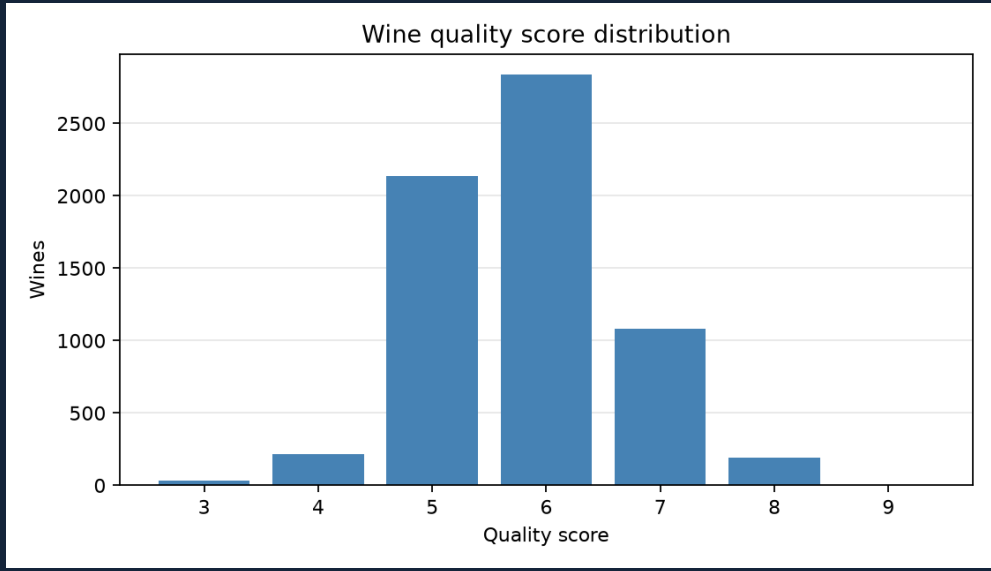
Final project | 30 October 2026

Target: quality score | UCI red and white Vinho Verde data

The dataset contains 6,497 wine samples

Two files, one regression target

- 1,599 red wines and 4,898 white wines
- 11 physicochemical predictors plus the quality score
- A wine_type input records which source file supplied each sample
- Zero missing cells in the original files
- 1,177 exact duplicate rows removed after combining



Quality scores are concentrated at 5 and 6. Score 9 has only five rows.

The test data stay out of model selection

One seeded split, then training-only cross-validation

Raw files

6,497 rows

Clean table

5,320 rows

80/20 split

4,256 train■1,064 test

CV + tuning

5 folds

- Seed 42 makes the split reproducible
- The final test set is evaluated after selecting the model

Duplicate removal and preprocessing are documented in the notebook and Python package.

Preprocessing is fitted inside each cross-validation fold

Numeric and categorical columns follow separate steps

Numeric predictors

- Median imputation
- StandardScaler
- 11 measurements

Wine type

- Most-frequent imputation
- OneHotEncoder
- red / white category

Each preprocessing branch sits in a scikit-learn Pipeline with its estimator.

This prevents validation and test rows from influencing fitted preprocessing values.

k-NN had the lowest five-fold training RMSE

The model was chosen using training data only

Selected settings: 21 neighbours, distance weighting. Ranking metric: mean CV RMSE.

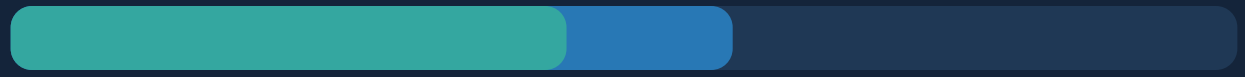
The selected model beat the mean baseline on the holdout

The same 1,064 test rows were used for both scores

RMSE

Dummy 0.868 | k-NN 0.685

MAE



Dummy 0.694 | k-NN 0.534

Test R^2: 0.375 | Test MAE: 0.534 score points

R^2 indicates that substantial score variation remains unexplained.

Alcohol ranked highest in permutation importance

Importance reflects this model and dataset, not cause and effect

- Predictive association does not prove causality
- Correlated inputs can affect importance rankings

A useful improvement over a simple baseline

The model still has clear limits

- Test RMSE improved from 0.868 to 0.685
- Average absolute error was 0.534 quality points
- The dataset describes Portuguese Vinho Verde wines
- The target is ordinal but modelled as a numeric score
- Saved pipeline supports repeatable JSON predictions

Reproducibility: package, CLI, notebook, tests and saved joblib model.

Source: Cortez et al. (2009), UCI Wine Quality, DOI 10.24432/C56S3T.